

FILIPPO GIOVAGNINI

Mathematics PhD Candidate | Machine Learning Research | Transformers & Agent Systems
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RESEARCH PROFILE

Mathematics PhD candidate at Imperial College London with research in deep-learning theory, high-dimensional probability, and stochastic analysis, alongside applied ML experience in NLP, transformer architecture, model evaluation, and unstructured-data pipelines. Strong Python and hands-on PyTorch experience, with demonstrated research writing and end-to-end ownership in an agentic ML competition.

SELECTED MACHINE LEARNING WORK

- **Kaggle Silver Medal – AI-agent and ONNX optimization.** Ranked 40th (top 1.4%) in *The 2026 NeuroGolf Championship*. Built an AI-agent workflow to propose, test, and refine compact ONNX solvers for ARC-AGI transformations under exact-output, parameter, and memory constraints. Click [here](#) to see our silver solution.
- **Decoder-only Transformer from scratch – PyTorch (ongoing).** Implemented a compact language model from first principles, including token embeddings, causal multi-head self-attention, residual blocks, normalization, and the language-model head; developing the TinyStories tokenization, training, and evaluation pipeline.

MACHINE LEARNING EXPERIENCE

Stripe Partners

Data Scientist, Part-time

London, UK

Mar 2026 – Present

- Develop transformer-based statistical pipelines, producing model-ready features and cross-market analyses for customers (including PayPal/Venmo, Novo Nordisk).
- Contribute to internal research on large-scale conversation data analysis. Click [here](#) to see out latest work on how users employ conversational AI in financial judgement and decision-making.

Aleta Index

Machine Learning Engineer, Part-time

London, UK

Feb 2025 – Feb 2026

- Built real-time NLP and data pipelines for topic, sentiment, and framing analysis across heterogeneous financial-news streams, transforming unstructured text into structured signals for monitoring and modeling.

SELECTED RESEARCH PUBLICATIONS

Go to my website for a complete list.

- **Universality in Deep Neural Networks: An approach via the Lindeberg exchange principle.** With S. Kotitsas and M. Romito, 2026. Quantitative Gaussian-process universality for fully connected neural networks with general random weights, using layerwise Lindeberg replacement and Wasserstein analysis.
- **Non-selection of Lagrangian trajectories in the zero-noise limit for a class of stochastic regularizations.** With L. Galeati and M. Sorella, 2026.
- **A uniform particle approximation to the Navier–Stokes–alpha models in three dimensions with advection noise.** With D. Crisan, 2025.

EDUCATION

Imperial College London

PhD in Mathematics – Stochastic Analysis, MFC CDT

London, UK

Dec 2023 – Present

University of Pisa

MSc in Mathematics – 110/110 cum laude, GPA 30/30

Pisa, Italy

2023

ETH Zürich

Exchange Programme – Quantitative Risk Management and Optimal Transport

Zürich, Switzerland

2023

University of Pisa

Bsc in Mathematics – 110/110 cum laude, GPA 29.85/30

Pisa, Italy

2023

TECHNICAL SKILLS

Programming & frameworks: Python, PyTorch, Scikit-learn, NumPy, Pandas, Polars, Git, Linux, C++.

Machine learning: Transformers and attention, NLP, embeddings, agentic workflows, model compression, evaluation and benchmarking, probabilistic modelling, feature engineering, real-time and unstructured-data pipelines.

Research methods: deep-learning theory, high-dimensional probability, stochastic calculus, SDEs/SPDEs, Monte Carlo methods, numerical analysis, statistical learning, optimal transport.

Additional: Teaching Assistant for MSc Stochastic Calculus for Finance at Imperial College Business School (2023–Present); Italian (native), English (fluent), Spanish (conversational).